THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET A

Chapter 2: **Acids, Bases and Salts** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The acid present in our stomach (gastric juice) is: 1

(a) citric acid **(b)** sulphuric acid
(c) acetic acid **(d)** hydrochloric acid

2. A solution turns red litmus blue. Its pH is likely to be: 1

(a) 7 **(b)** 1 **(c)** 4 **(d)** 10

3. Bleaching powder is prepared by the action of chlorine on: 1

(a) dry slaked lime $Ca(OH)_2$ **(b)** $CaCO_3$
(c) $NaOH$ **(d)** Na_2CO_3

4. A student states: “All bases are alkalis.” The correct response is: 1

(a) Correct
(b) Incorrect — only water-soluble bases are alkalis
(c) Correct for metal oxides
(d) Cannot be determined

5. Student A says the pH of pure water is 7; Student B says 0. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

6. The chemical formula of washing soda is: 1

(a) $NaHCO_3$ **(b)** $CaSO_4 \cdot 2\,H_2O$ **(c)** $Na_2CO_3 \cdot 10\,H_2O$ **(d)** Na_2CO_3

7. When zinc reacts with sodium hydroxide, the gas evolved is: 1

(a) carbon dioxide **(b)** chlorine **(c)** hydrogen **(d)** oxygen

8. Tooth decay starts when the pH of the mouth is lower than: 1

(a) 7.5 **(b)** 3 **(c)** 5.5 **(d)** 9

9. The salt used as an antacid ingredient and in baking powder is: 1

(a) calcium sulphate **(b)** sodium chloride
(c) sodium carbonate **(d)** sodium hydrogencarbonate

10. The ions responsible for the acidic nature of an aqueous solution are: 1

(a) H_3O^+ **(b)** Cl^- **(c)** OH^- **(d)** Na^+

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why does dry HCl gas not change the colour of dry blue litmus paper? 2

12. Write the chemical name and formula of Plaster of Paris. 2

13. Why is it advised not to keep curd and sour substances in brass or copper vessels? 2

14. A student says that diluting an acid increases its H_3O^+ concentration per unit volume. Justify or refute. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. What is a neutralisation reaction? Give an equation, and one example of neutralisation in daily life. 3

16. How is baking soda prepared? Write the equation and two of its uses. 3

17. Five solutions A–E have pH 4, 1, 11, 7 and 9. Identify 3
(i) neutral
(ii) strongly alkaline
(iii) strongly acidic
(iv) weakly acidic
(v) weakly alkaline. Arrange them in increasing order of H^+ concentration.

OR

Explain the chlor-alkali process with the equation and name the products formed at the anode and cathode. 3

18. What is water of crystallisation? Write the formula of blue vitriol and explain what happens when it is heated. 3

OR

How does the flow of acid rain water into a river make the survival of aquatic life difficult? What is the pH range for human body and aquatic life? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) What happens when an acid reacts with a metal carbonate? Write the equation for Na_2CO_3 and HCl. 5
(ii) How will you test the gas evolved? Write the equation.
(iii) What happens on passing excess of this gas?

OR

(i) How is Plaster of Paris obtained? Write the equation. 5
(ii) What happens when it is mixed with water?
(iii) Give two uses.
(iv) Why should it be stored in a moisture-proof container?

20. **Case study:** Neha was stung by a honey-bee. Her mother applied baking soda on the sting. 5
(i) Name the acid injected by the bee.
(ii) Why was baking soda applied?
(iii) Name the type of reaction.
(iv) What is the pH of baking soda solution — less or more than 7?
(v) Suggest one other remedy.

OR

(i) What is meant by strong and weak acids? Give one example of each. 5
(ii) What is a universal indicator?
(iii) Explain the importance of pH in
(a) digestion
(b) soil.

END OF QUESTION PAPER **Total: 40 Marks**